

Whole-Body Physics Signatures for Humanoid Gait Failure Ranking

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Abstract—Reduced-order checks omit whole-body constraints that can invalidate humanoid gait candidates. We test whether normalized whole-body physics signatures add offline failure-ranking information beyond gait parameters and robot identity on TALOS and iCub. Fixed histogram-based gradient-boosting (HGB) models are fit on 200 development rollouts and evaluated without refitting on two independent 400-rollout scrambled-Sobol confirmations (seeds 42026 and 52026). The whole-body model reaches pooled failure PR-AUC/ROC-AUC of .725/.770 versus .689/.763 for the fair parameter-plus-robot comparator. Its prespecified macro within-robot PR-AUC difference is +.034, with a conditional 95% bootstrap interval [-.009,.073]. The direction is positive on both scrambles (+.017 and +.032), but the interval crosses zero, so the confirmatory gate fails. Phase-resolved and phase-agnostic variants are tied, and a rollout-budget diagnostic shows no efficiency advantage. The evidence suggests a modest whole-body ranking signal but does not establish a reliable improvement.

Keywords: humanoid locomotion; whole-body dynamics; failure ranking; feasibility screening; reproducible simulation

I. INTRODUCTION

Reduced-order locomotion methods make candidate generation tractable through preview control, capture-point reasoning, footstep placement, and divergent components of motion [1], [2], [3], [4]. Their candidates can nevertheless fail once whole-body dynamics, unilateral contacts, actuator limits, timing error, and impacts are considered. Because calling a whole-body rollout oracle for every offline candidate is costly, a preliminary ranking score could reduce the number of required evaluations. Learning feasibility constraints [5] and learning from costly gait trials [6] motivate this direction, but neither line of work eliminates the need for independent evaluation across robot platforms.

We therefore ask a narrower, directly testable question: after all engineering choices are frozen, do normalized whole-body physics features add repeatable failure-ranking information beyond gait parameters and robot identity on new TALOS [7] and iCub [8] rollouts? The robots remain separate evaluation strata; a pooled gain is not interpreted as morphology transfer.

Our contributions are:

- 1) a normalized whole-body sequence signature evaluated by a separate closed-loop rollout oracle;
- 2) two independently scrambled 400-rollout confirmations with a frozen macro within-robot endpoint; and

- 3) a fair robot-aware comparator, representation ablations, and an honest negative result when the conditional interval includes zero.

Two 100-row development sets determine the models. Two separate 400-row confirmation sets are then used once for ranking, conditional bootstrap inference, and the budget diagnostic; their labels never alter the models or oracle.

II. RELATED WORK

A. Whole-Body Locomotion Feasibility

Optimization-based locomotion ranges from hierarchical inverse kinematics to centroidal and multicontact optimal control [9], [10], [11], [12], [13]. Timing adaptation and perceptive nonlinear MPC further show why contacts and execution timing matter beyond a fixed LIPM plan [14], [15]. We focus on offline failure ranking and ask whether inexpensive features derived from a candidate sequence can rank failures before a separate whole-body rollout.

Carpentier *et al.* learned whole-body kinematic feasibility constraints for multicontact planning [5]. Wrench polytopes and improved feasible regions combine contact, actuation, dynamic-balance, and kinematic limits [16], [17]. Sequence-level prediction can also reorder costly planning attempts [18]. Our contribution lies in the experimental design: the score summarizes an entire perturbed gait, the label comes from a separate closed-loop rollout, and the same frozen representation is evaluated on two humanoid models and two independent scrambles. The contribution is the ranking representation and evaluation protocol, not a new controller or outer-loop search algorithm.

B. Ranking, Calibration, and Selective Risk

Ranking metrics characterize discrimination, whereas selective classification characterizes risk conditional on admission [19], [20]. Probability calibration, conformal prediction, and risk-controlling prediction offer principled tools when an independent calibration sample and exchangeability assumptions are available [21], [22], [23], [24]. Because confirmation has no separate calibration split, we report ranking and rollout-budget behavior without a calibrated-risk guarantee. Discrimination alone does not certify selective safety.

III. RANKING SCORES AND FROZEN TRAINING DATA

A. Sequence Representation and Oracle

Each six-step candidate is normalized by neutral leg length L , neutral width w_0 , and $T = \sqrt{L/g}$. Endpoint- C^2 foot trajectories are generated with warm-started bounded manifold inverse kinematics, using manifold integration and differencing rather than Euclidean quaternion arithmetic. At each reference state, a two-stage linear program solves for generalized acceleration a , joint effort τ , and world-aligned sole wrenches λ subject to

$$M(q)a + h(q, v) = S^\top \tau + J_c^\top \lambda, \quad (1)$$

rigid-contact acceleration equalities, effort bounds, unilateral normal force, a friction pyramid, torsional friction, and convex-polygon CoP bounds. The primary objective minimizes normalized L_1 acceleration-tracking slack; at that optimum, the secondary objective minimizes normalized effort and wrench magnitude. The LP produces ranking features; it does not define the rollout label.

For each active sole, the equality $J_c a + \dot{J}_c v = 0$ is imposed in a world-aligned contact frame. With sole half-length ℓ , half-width b , and wrench $\lambda = (f_x, f_y, f_z, m_x, m_y, m_z)$, the inequality block includes

$$\begin{aligned} f_z &\geq 0, & |f_x|, |f_y| &\leq \mu f_z, \\ |m_x| &\leq b f_z, & |m_y| &\leq \ell f_z, \end{aligned} \quad (2)$$

plus the torsional bound and effort limits. Slack is attached to desired acceleration rather than rigid-body consistency, so the dynamics equality remains auditable. A lexicographic second solve prevents arbitrary large opposing wrenches among equally good acceleration fits.

Signed channels encode torque, friction, CoP/ZMP, joint-position, and joint-velocity margins. Nonnegative channels encode IK residual, acceleration-tracking slack, solver failure, and a touchdown proxy. The touchdown proxy combines normalized pre-impact contact speed with disturbance magnitude; it is not an estimated impact impulse. For single support, touchdown, and double support separately, every channel is aggregated by minimum, fifth percentile, median, 95th percentile, and maximum. All signature channels are normalized by robot capacities. For the proposed B5, the three phase values for each channel/statistic are sorted, preserving their values and dimension while removing phase identity. The phase-resolved representation is reported as an ablation rather than claimed as the source of improvement.

Formally, let $c_j(t)$ denote channel j and $\mathcal{I}_p$ the samples in phase p , for $p \in \{\text{SS}, \text{TD}, \text{DS}\}$. Using $\mathcal{A} = \{\min, Q_{.05}, Q_{.50}, Q_{.95}, \max\}$, the signature is

$$\phi = [a(\{c_j(t) : t \in \mathcal{I}_p\})]_{p,j,a \in \mathcal{A}}. \quad (3)$$

Ten channels, three phases, and five aggregations yield 150 values. Appending the eleven gait/disturbance parameters and one binary robot indicator gives B5 a 162-dimensional input; B4 receives the same parameters and robot indicator. Thus their difference isolates summarized whole-body physics rather than classifier capacity or unequal access to robot identity.

A separate closed-loop whole-body rollout oracle, implemented in Pinocchio [25], provides labels for both robots. Continuous single/double support uses constrained rigid-body dynamics; touchdown first tests admissible perfectly inelastic sticking-contact subsets and otherwise falls back to a unilateral vertex-contact solve. Feed-forward inverse dynamics plus normalized PD feedback controls the nominal model. Payload changes only the torso inertia in the plant model, while friction variation, timing error, and a seeded horizontal impulse perturb execution. The labels therefore reflect closed-loop model mismatch rather than a second evaluation of the reference generator.

With $\omega = \sqrt{g/L}$, the controller requests

$$a_d = a_{\text{ref}} + 20\omega^2 \text{ difference}(q, q_{\text{ref}}) + 8\omega(v_{\text{ref}} - v). \quad (4)$$

Inverse dynamics maps a_d to feed-forward-plus-feedback effort under the realized contact set. We log unconstrained demand, then resolve with URDF effort limits when demand saturates. The plant integrates the resulting constrained acceleration on the configuration manifold. At touchdown, admissible sticking subsets are tested with impulse dynamics; an infeasible full-sole impact may drop to unilateral vertex contacts rather than being accepted by construction.

The first-failure categories are normal-force, friction, CoP, joint position/velocity, collision, fall, tracking, and dynamics. Contact failure is evaluated from realized rollout wrenches, not a reduced-order ZMP formula. A rollout also fails if pelvis height falls below 65% of neutral, roll/pitch exceeds 30° , CoM error exceeds $0.12L$, or contact/impact dynamics cannot be solved. Applied torque is saturated at URDF limits; the unsaturated demand is diagnostic rather than a failure label. Joint position or velocity must exceed the corresponding limit by more than 1% for at least 20 ms. These rigid-contact mechanics and numerical transition tolerances are evaluation choices, not a physical safety certificate.

The signature and label deliberately use different information. Signature channels are computed along the planned whole-body reference by the inverse-dynamics LP. The rollout begins from that reference but evolves a payload-perturbed plant under saturated feedback, realized contact transitions, and a seeded impulse. Consequently, a large signature margin may coexist with a failed rollout, and B5 must learn associations rather than reproduce an oracle rule. This separation prevents target leakage from realized failure times or rollout wrenches into the ranking features.

B. Normalization and Oracle Validity

Geometric inputs use robot-specific physical capacities. Lengths are divided by L or the relevant sole half-dimension, durations by T , forces by mg , joint efforts by URDF effort limits, and external impulse by $m\sqrt{gL}$. The neutral width w_0 is the lateral distance between sole centers. Sole support polygons are the convex hulls of the lowest vertices in the manually reviewed, axis-normalized collision meshes. This design aims to make equivalent margins comparable. Both learned scores also receive the same binary robot indicator so

the comparator is not denied base-rate information available implicitly in physical features.

Several numerical guards prevent transition artifacts from causing failures. The CoP moment inequality includes both a $10^{-4}L$ geometric tolerance and an absolute force-tolerance floor, so a nearly unloaded contact is not divided by an arbitrarily small normal force. Flat-terrain penetration has a $10^{-3}L$ continuous limit and a $2 \times 10^{-3}L$ hard transition bound. Within that band, contact state, vertical velocity, and one-step predicted gap determine whether the event is recovery, liftoff, or reimpact. Scheduled and realized contacts are logged separately; the oracle does not force a planned double-support phase to remain active.

Before confirmation, the unchanged oracle passed the repository’s analytic validity suite: valid manifold configurations and unit quaternions, normalized rigid-body residual below 10^{-6} , reviewed wrench/CoP signs and frames, a feasible static double-support pose, and correct rejection of injected torque, friction, collision, and impact failures. No oracle parameters were changed for the evaluation. Each case is simulated at 100 Hz and written atomically before aggregate statistics are updated.

C. Five Fixed Ranking Scores

The two development sets contain 200 labeled rollouts, exactly 100 per robot. They include all oracle and representation engineering; they are used to freeze the proposed representation and fit B4/B5. The HGB configuration uses a maximum of seven leaf nodes, learning rate 0.08, 150 iterations, ℓ_2 regularization 1.0, and random state 2026; there is no tuning, threshold selection, recalibration, or refitting with confirmation outcomes. We compare five scores:

- B1 ZMP/CoP margin;
- B2 IK and joint-limit margin;
- B3 worst phase-aggregated dynamics slack;
- B4 HGB using gait/perturbation parameters and robot identity; and
- B5 HGB using the same inputs plus the normalized, phase-agnostic whole-body signature.

For all scores, a larger value means a higher predicted likelihood of oracle failure. The physics baselines are deliberately simple. Giving the same robot indicator to B4 and B5 prevents morphology-dependent failure prevalence from being credited only to B5. HGB is a standard compact nonlinear baseline [26].

IV. INDEPENDENT CONFIRMATION PROTOCOL

The revised endpoint and code were frozen before evaluation (commit 88c1b41). An earlier seed-32026 set informed this revision and is excluded from all results below. Confirmations use seeds 42026 and 52026, each with 400 new six-step rollouts (200 per robot). Every rollout seed is disjoint from development and from the other confirmation. The frozen common domain is step length $[0.10, 0.16]L$, width $[1.00, 1.10]w_0$, single-support duration $[2.70, 2.80]T$, double-support duration $[0.70, 0.80]T$, CoM-height scale $[0.92, 0.98]$, ZMP biases $x \in [-0.05, 0.05]$

TABLE I
FROZEN DATA ROLES

Set	Seed	Samples/robot	Payload	Label use
Development A	12026	50	$[0, .10]m$	engineer/fit
Development B	22026	50	$[0, .05]m$	select/fit
Confirmation A	42026	200	$[0, .05]m$	test once
Confirmation B	52026	200	$[0, .05]m$	test once

and $y \in [-0.25, -0.10]$, friction $[0.4, 0.8]$, payload $[0, 0.05]m$, timing error $[-0.02, 0.02]$ s, and normalized horizontal impulse $[0, 0.04]$. Gait and perturbation marginals use separately scrambled Sobol streams with a deterministic pairing.

The primary endpoint is the macro within-robot difference

$$D = \frac{1}{2} \sum_{r \in \{\text{TALOS}, \text{iCub}\}} [\text{AP}_r(\text{B5}) - \text{AP}_r(\text{B4})], \quad (5)$$

where PR-AUC treats failure as positive. We draw 10,000 paired bootstrap samples within each (scramble, robot) stratum. The gate requires a positive percentile-CI lower endpoint and a positive point difference on each scramble. Confirmation labels cannot alter the oracle, domain, representation, models, hyperparameters, or seeds.

Scores are predicted once and held fixed across resamples. Because the rows come from two scrambled quasi-random designs rather than IID sampling, the interval is explicitly conditional and descriptive, not a nominal population coverage guarantee. Pairing compares B4/B5 on identical cases; the two independent scrambles directly expose sign stability.

As a descriptive deployment diagnostic, 10,000 deterministic pools of 50 confirmation candidates per robot are sampled without replacement. B4 and B5 rank the same pool from lowest predicted failure probability upward; we record success within budgets $\{5, 10, 20\}$ and the rank of the first feasible gait. The artifact verifies row counts, seed disjointness, source/protocol fingerprints, and SHA-256 hashes before analysis.

V. CONFIRMATION RESULTS

The frozen oracle returns 283/400 (70.8%) successful TALOS rollouts and 159/400 (39.8%) successful iCub rollouts. TALOS has 107 friction, seven normal-force, two tracking, and one impact-dynamics failure; iCub has 109 friction, 106 normal-force, ten impact-dynamics, eight tracking, seven collision, and one CoP failure. Mean oracle runtimes are 13.30 s and 9.94 s, respectively. The different base rates motivate within-robot inference.

We report PR-AUC because failure prevalence differs, with ROC-AUC as a complementary ranking metric [27]. Table II contains confirmation-only results.

B5 reaches pooled PR-AUC/ROC-AUC .725/.770 versus .689/.763 for the fair B4 comparator. The macro within-robot difference is +.034 with conditional 95% CI $[-.009, .073]$ (Fig. 1). Point differences are positive on both independent scrambles (+.017 and +.032), but the interval crosses zero; the prespecified gate therefore fails. Combined per-robot gains are modest on TALOS (.512 versus .495) and larger on iCub

TABLE II
CONFIRMATION FAILURE-RANKING METRICS

Score	PR-AUC	ROC-AUC	TALOS PR	iCub PR
B1 ZMP/CoP	.596	.659	.300	.653
B2 IK/joint	.574	.655	.302	.603
B3 slack	.630	.675	.273	.694
B4 parameters+ID	.689	.763	.495	.738
B5 whole-body	.725	.770	.512	.789

Macro B5 - B4 = +0.034 (conditional 95% CI -0.009, +0.073)

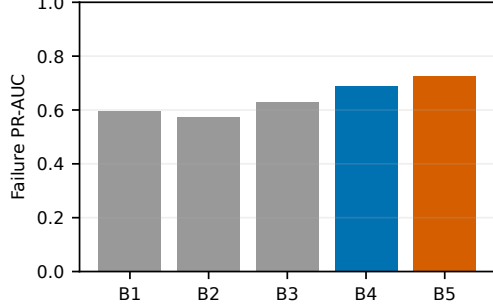


Fig. 1. Confirmation PR-AUC. The prespecified macro within-robot B5–B4 difference is positive, but its conditional bootstrap CI includes zero.

(.789 versus .738). Relative to the strongest simple margin B3, B5 adds .095 pooled PR-AUC, but this secondary point comparison has no confirmatory interval.

A. Representation and Rollout Budget

Table III tests the central representation choices. Phase-resolved and phase-agnostic variants are effectively tied (.727 versus .725 pooled PR-AUC); removing touchdown features gives .712. Thus the experiment does not support a benefit from phase identity.

At budget five, both scores find a feasible gait in at least 99.89% of 50-candidate pools; all budget-10 and budget-20 trials succeed. Mean ranks to the first feasible gait are 1.000 (B4) versus 1.011 (B5) for TALOS and 1.074 versus 1.319 for iCub (Fig. 2). Thus the whole-body score does not reduce rollout cost in this high-coverage domain.

VI. DISCUSSION

The repeated confirmation gives a deliberately narrower result than the exploratory analysis. Whole-body features show the same positive point direction on two new scrambles, but the macro effect is small and its conditional interval includes zero. The evidence therefore suggests additional ranking information but does not establish a reliable improvement over a fair parameter-plus-robot comparator. The larger iCub gain and small TALOS gain also show that one pooled number should not be read as morphology invariance.

Phase identity is not supported: phase-resolved and phase-agnostic representations are tied, while touchdown removal is mildly worse only in pooled data. Feasible candidates are also common enough that both learned scores almost always

TABLE III
CONFIRMATION REPRESENTATION ABLATION

Representation	Pooled PR/ROC	TALOS PR/ROC	iCub PR/ROC
Phase-agnostic (B5)	.725/.770	.512/.708	.789/.736
Phase-resolved	.727/.772	.522/.720	.790/.734
No touchdown	.712/.770	.536/.718	.765/.715

succeed within five rollouts; improved discrimination does not imply lower search cost in this domain.

The apparent tension is expected. PR-AUC measures ordering over all failures, including difficult cases deep in the list. Search success at a small budget depends only on whether at least one of the first few candidates is feasible. With 70.8% TALOS and 39.8% iCub feasibility, that event saturates even when the detailed failure ordering improves. These metrics answer different questions and should not be substituted for one another.

For use, the workflow is therefore score, shortlist, then verify every selected candidate with the full oracle. A shared acceptance rule would require new calibration data and a prespecified risk-control procedure. The score is a ranking aid, not a safety filter.

VII. REPRODUCIBILITY AND LIMITATIONS

Supplementary Artifact S1 archives both development CSVs, all 800 confirmation records from two scrambles, source/environment fingerprints, validity reports, fixed analysis, and figures. Supplementary Video V1 shows successful and failed rollouts for both robots. One command refits B4/B5, verifies hashes and seed disjointness, and recreates the reported JSON and figures. Robot meshes are not redistributed. The same code and data are available in the public repository [28].

Without a calibration split, the study cannot evaluate 5% risk control via a selective threshold or Clopper–Pearson bound, or make an exchangeability-based guarantee. The budget analysis reuses confirmation candidates and is descriptive. All labels come from one simulator stack, and the rigid-contact oracle is not a physical safety certificate. The scope is two simulated morphologies on flat ground, six-step near-neutral references, one controller, and one reviewed oracle. Compliance, state estimation, latency, thermal effects, hardware contacts, and leave-one-morphology-out transfer remain outside the evidence.

VIII. CONCLUSION

Across 800 untouched TALOS/iCub rollouts in two independent scrambles, the whole-body model reaches .725 pooled PR-AUC versus .689 for a fair parameter-plus-robot comparator. The macro within-robot difference is +.034 with conditional 95% CI [-.009,.073]: positive on both scrambles, but inconclusive because the interval crosses zero. Phase identity and rollout-budget savings are not supported. The result is an honest simulation benchmark, not evidence of calibrated risk control, transfer, or safety.

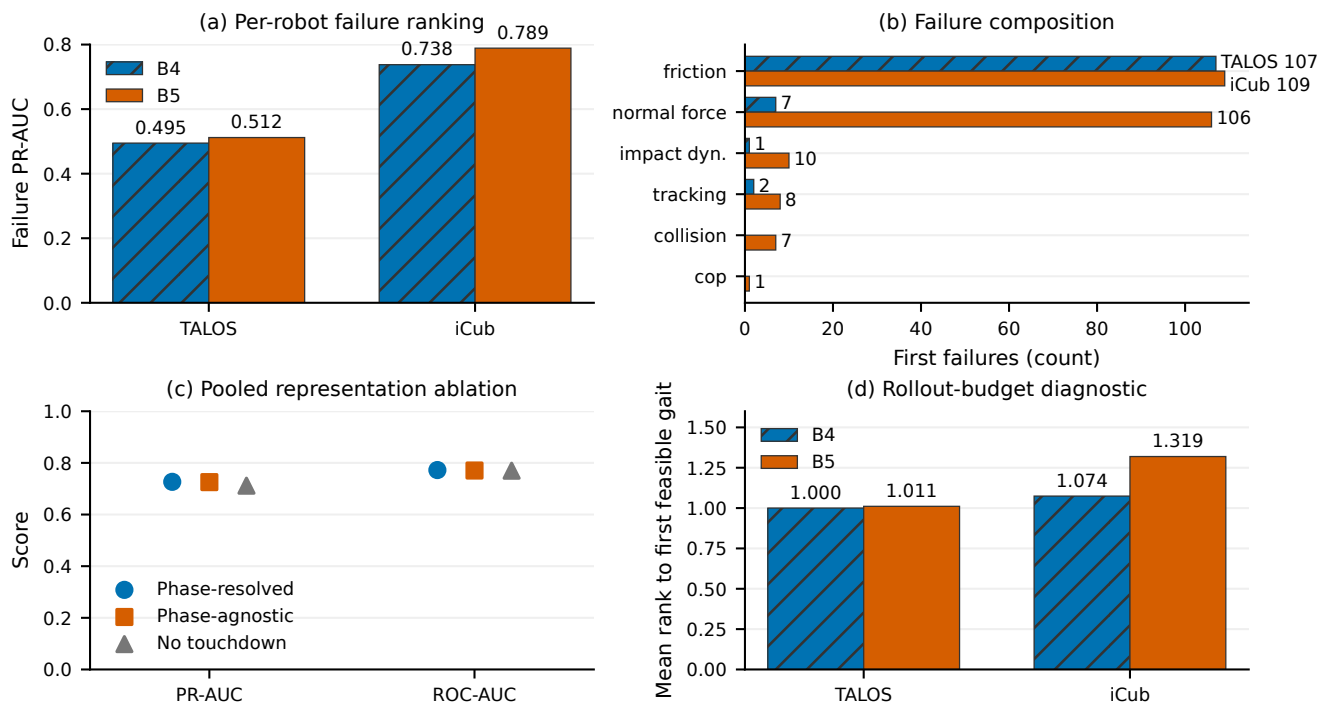


Fig. 2. Repeated-confirmation summary. (a) B5 has a small positive PR-AUC point difference on both robots. (b) First-failure composition differs by robot. (c) Phase-resolved and phase-agnostic variants are tied. (d) B5 yields no rollout-budget advantage (lower is better).

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DECLARATION OF COMPETING INTEREST

The author declares that he has no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

DATA AND CODE AVAILABILITY

The development and confirmation records, analysis and figure-generation code, environment and source fingerprints, validity reports, and rollout video are available in Supplementary Artifact S1 and the public repository [28].

CREDIT AUTHORSHIP CONTRIBUTION STATEMENT

Egor Zharinov: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing—original draft, Writing—review and editing, Visualization.

USE OF GENERATIVE AI

OpenAI Codex assisted with research-code implementation, review, and language clarity. The author verified all outputs and takes full responsibility for the article.

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